

Solution Requirements

Date	02 July 2026
Team ID	SWTID-2026-9794
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>The system shall allow applicants and administrators to securely log in using email/mobile number and password with OTP verification for sensitive actions.</i>	High
2	Authorization levels	<i>The system shall provide role-based access where Applicants can submit and track applications, while Administrators can verify applications, manage users, and view prediction reports.</i>	High

3	External interfaces	<i>The system shall integrate with Credit Bureau APIs, Aadhaar/PAN verification services, bank databases, and email/SMS notification services for applicant verification and communication.</i>	High
4	Transactions processing	<i>The system shall receive applicant information, validate data, preprocess financial details, execute the machine learning prediction model, and generate approval or rejection results in real time.</i>	High
5	Reporting	<i>The system shall generate applicant status reports, approval/rejection reports, prediction accuracy reports, dashboards, and downloadable PDF/Excel reports for administrators.</i>	Medium
6	Business rules, etc.	<i>Credit card approval shall be determined using predefined eligibility rules such as minimum income, credit score, debt-to-income ratio, employment status, and ML prediction score.</i>	High
7	Compliance to laws or regulations	<i>The system shall comply with data privacy regulations, secure storage of personal information, banking security standards, and obtain user consent before processing personal data.</i>	High

8	<i>Other</i>	<i>The system shall send automatic email/SMS notifications regarding application submission, verification progress, approval, or rejection status.</i>	Medium
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Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>The system shall process applicant data and display the credit card approval prediction quickly with minimal response time.</i>	<i>Prediction result displayed within 3 seconds for 95% of requests.</i>
2	Scalability	<i>The system shall efficiently support an increasing number of users and applications without performance degradation</i>	<i>Supports at least 10,000 concurrent users and processes 50,000+ applications per day.</i>
3	Security & Data Privacy	<i>The system shall securely protect applicant data using encryption, authentication, and secure communication protocols.</i>	<i>AES-256 encryption for stored data, HTTPS/TLS for data transmission, Role-Based Access Control (RBAC) enabled.</i>
4	Reliability & Availability	<i>The system shall remain operational with minimal downtime and provide reliable prediction results.</i>	<i>(e.g., 99.9% uptime per month) 99.9% monthly uptime, automatic backup every 24 hours, recovery within 15 minutes after failure.</i>

5	Usability & Accessibility	<i>The system shall provide an intuitive, responsive, and user-friendly interface accessible across multiple devices and browsers.</i>	<i>Responsive design, compatible with major browsers, and follows WCAG 2.1 AA accessibility guidelines.</i>
6	Other	<i>The system shall be modular, easy to maintain, support future machine learning model updates, and be deployable on different platforms.</i>	Modular architecture, deployment on Windows, Linux, and Cloud platforms, and ML model updates without affecting existing functionality.